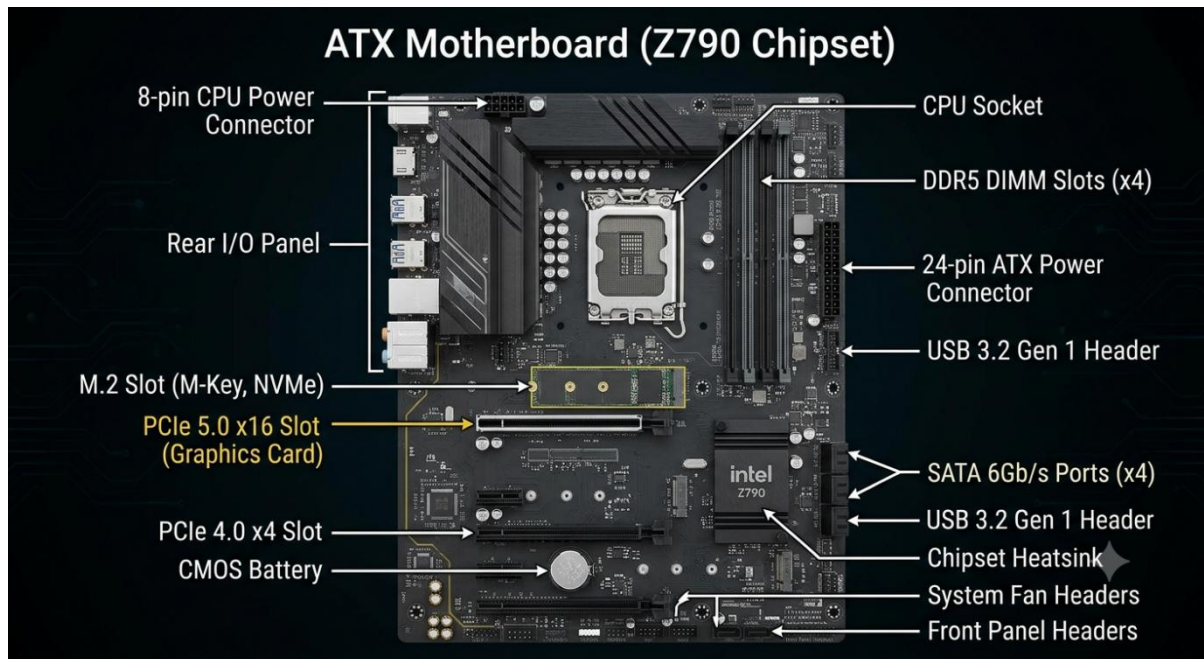


1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

1. Socket AM4 (AMD)

- **Compatible CPU Model:** AMD Ryzen 7 5800X3D or AMD Ryzen 5 5600X
- **Overview:** Introduced by AMD in 2016, AM4 supported multiple generations of Ryzen processors using DDR4 memory.

2. Socket AM5 (AMD)

- **Compatible CPU Model:** AMD Ryzen 7 7800X3D or AMD Ryzen 9 9950X
- **Overview:** AMD's modern platform introduced with the Ryzen 7000 series, featuring native DDR5 memory and PCIe 5.0 support.

3. LGA 1700 (Intel)

- **Compatible CPU Model:** Intel Core i7-13700K or Intel Core i9-14900K
- **Overview:** Intel's desktop socket introduced with 12th Gen processors, also supporting 13th and 14th Gen Core series CPUs.

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU (Central Processing Unit):

The CPU is the "brain" of the computer. It processes instructions, performs

calculations, and controls the operation of programs. When you open an app like Spotify, the CPU loads the program into memory, executes its instructions, and manages tasks such as displaying the interface and processing your actions.

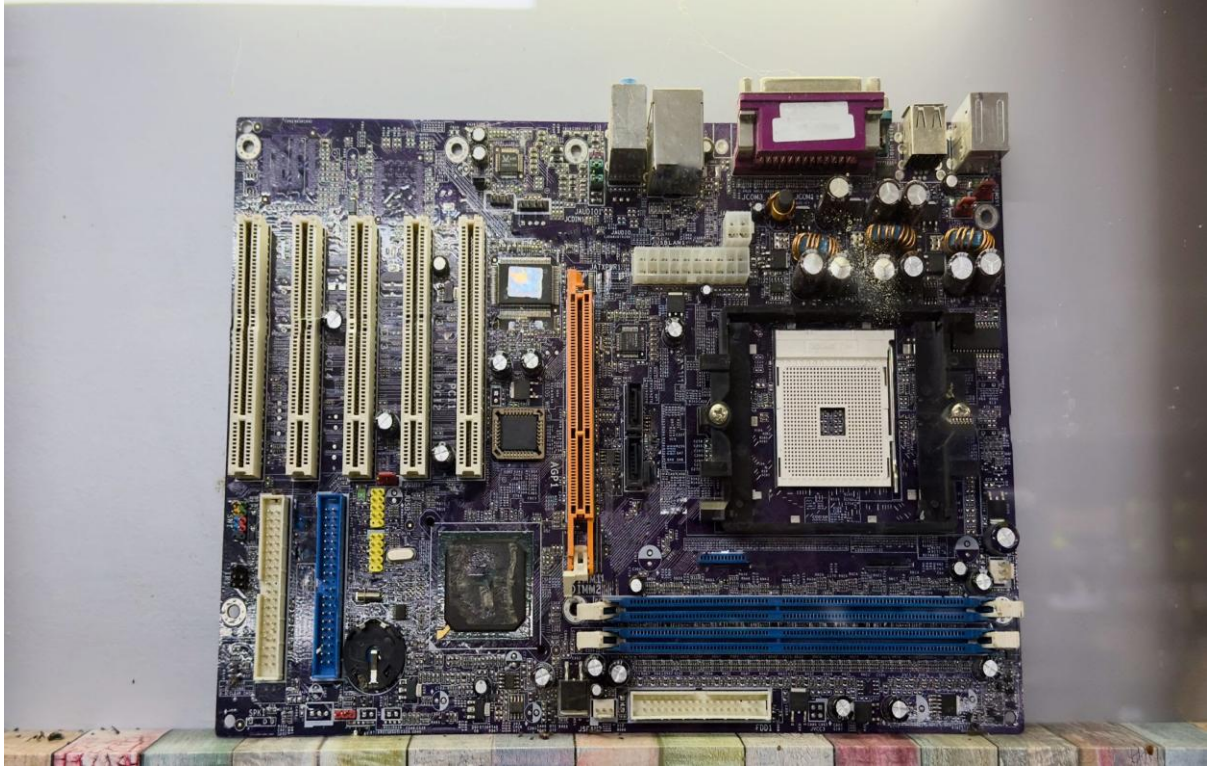
Chipset:

The chipset is a group of controllers on the motherboard that manages communication between the CPU and other hardware components, such as the RAM, storage drive, USB ports, graphics, and audio devices. It helps ensure that data moves efficiently between these parts.

How they work together when opening Spotify:

When you click the Spotify icon, the chipset retrieves the program data from the storage drive and sends it to the RAM. The CPU then processes the instructions to launch the app. As Spotify runs, the chipset continues to handle communication between the CPU, memory, storage, internet connection, speakers, and other devices, while the CPU processes your commands, such as searching for songs or playing music. Together, the CPU and chipset ensure the application runs smoothly and responds quickly to your actions.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one



1. CPU Socket (Processor Socket)

- Location: Center-right region of the motherboard (large square socket with a lever arm).
- Function: Houses the CPU. It contains thousands of tiny pin contacts or holes to allow high-speed communication between the processor and the rest of the motherboard.

2. RAM Slots (DIMM Slots)

- Location: Long vertical slots located to the right of the CPU socket (shown in blue/beige with locking clips at the ends).

- Function: Holds Dual In-line Memory Modules. These supply the high-speed temporary memory your system needs to run active applications and processes.

3. PCIe Expansion Slots (PCI Express)

- Location: the long orange slot for GPUs, along with white/beige legacy PCI/PCIe slots.
- Function: Used to plug in expansion cards such as dedicated graphics cards (GPUs), Wi-Fi adapters, sound cards, and capture cards.

4. Rear I/O Ports Panel

- Location: Top-left edge of the board (facing the exterior back panel of the computer case).
- Function: Provides external connectivity for peripherals:
 - USB Ports: Connect keyboards, mice, and external drives.
 - Ethernet (RJ-45) Port: Connects to wired local networks and the internet.
 - Audio Jacks: Output and input for speakers, headphones, and microphones.
 - Video Outputs (HDMI / DisplayPort / VGA): Send display signals directly from integrated graphics to a monitor.

5. Storage Connectors (SATA / IDE Ports)

- Location: Bottom-left/bottom-center area blue and yellow rectangular connectors.

- Function: Connect internal storage drives SATA HDDs, SSDs, and optical drives via data cables to transport files and operating system data.